Boulos Chaaya

Full-Stack Developer · TypeScript, Node.js, React, PostgreSQL

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PROFILE

Full-stack developer who builds production-grade web systems end to end, from data model and API to interface, CI/CD, containers and monitoring. Works primarily in TypeScript, React/Next.js, Node.js and PostgreSQL. Focused on correctness under failure: transactional state, concurrency control, and automated tests that run against real databases and simulate outages.

PROJECTS

ProjectOps — Client project management platform 2026

Node.js · TypeScript · PostgreSQL · React 19 · Vite · GitHub Actions · Playwright · Render Live demo · GitHub

- Built a complete project lifecycle application, covering intake, estimates, planning, delivery, invoicing and acceptance, with a full audit trail.
- Engineered the backend on Node's native HTTP server with hand-written parameterized SQL (single runtime dependency); React 19 client served same-origin under a strict CSP.
- Implemented revision-based optimistic concurrency, proven by race-condition tests against real PostgreSQL that allow exactly one winner.
- Introduced keyset pagination, reducing a 20,000-record list request from 208 ms / 4 MB to 7 ms / 10 KB.
- Delivered a public demo that gives each visitor an isolated, auto-expiring sandbox in its own PostgreSQL schema.
- Automated CI/CD with post-deploy verification and 15-minute uptime checks; every page tested for mobile layout and WCAG 2.1 AA.

Distributed Engine — Multi-tenant task and workflow execution engine 2026

TypeScript · Node.js · PostgreSQL · Redis/BullMQ · Next.js · Docker · Kubernetes · Istio · OpenTelemetry GitHub

- Designed PostgreSQL as the source of truth with a transactional outbox; a reconciler rebuilds the Redis queue after data loss.
- Implemented leased task claims with stalled-worker recovery, verified by a CI chaos test that kills a worker and flushes Redis mid-run with no lost or duplicated results.
- Built DAG workflow execution (fan-out / fan-in) with transactional step release and failure propagation limited to downstream steps.
- Added multi-tenancy (quotas, rate limiting, weighted fair scheduling, API-key provisioning) and cron scheduling with misfire and overlap policies.
- Deployed to Kubernetes with Istio ambient mTLS in CI, where a runtime verifier asserts both allowed and denied network paths.
- Covered by 490 automated tests against real PostgreSQL and Redis; secured operator dashboard behind a server-side proxy.

Portfolio — Personal website with content management dashboard 2026

Next.js 16 · TypeScript · Tailwind CSS · Drizzle ORM · PostgreSQL (Neon) · Auth.js · Resend · Vercel Live site · GitHub

- Built a site for projects, experience, blog and contact, managed through a private single-admin dashboard.
- Enforced authorization at the data-access layer for every read, server action and export, with tests for unauthenticated access.
- Implemented validated server actions with explicit error reporting, database-backed rate limiting and on-demand page regeneration.
- Ran end-to-end Playwright tests against PostgreSQL on every pull request.

EXPERIENCE

Web Developer (Freelance) — ZadAgency, Beirut 2024

- Designed, developed and deployed the agency website end to end, from requirements to launch.
- Implemented the service and portfolio showcases and a contact workflow using Resend.

IT & Support — DEP (Design Engineering Partner), Antelias 2018 – 2019

- Developed Excel VBA automation for internal workflows.
- Provided Windows, networking and infrastructure support, and resolved end-user technical issues.

TECHNICAL SKILLS

Frontend	TypeScript, JavaScript, React, Next.js, Tailwind CSS, shadcn/ui, Vite
Backend	Node.js, Express, Hono, tRPC, REST APIs, BullMQ
Data	PostgreSQL (SQL, transactions, concurrency control), Redis, MySQL, MongoDB, Drizzle ORM, Prisma
DevOps & cloud	Docker, Kubernetes, Istio, GitHub Actions, Linux, Git, Vercel, Render, Neon
Observability	OpenTelemetry, Grafana, Loki, structured logging
Testing	Jest, Playwright, integration testing against real databases, chaos testing
Services	Auth.js, Firebase, Supabase, Nhost, Stripe, Resend

EDUCATION

Information Technology, Programming Track (LT)
Educational Technical Institute, Dekwaneh · 2022 – 2023

Information Technology (TS1 – TS2)
Tourism Institute, Dekwaneh · 2016 – 2018

LANGUAGES

Arabic — Native

English — Professional working proficiency

French — Reading and writing